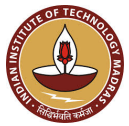
D'Alembert's Solutions

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The Cauchy Problem for an Infinite String

- Consider transverse vibrations of a very long (infinite) string:

$$u_{tt} = k^2 u_{xx}, \qquad x \in \mathbb{R}, \ t > 0. \tag{1}$$

- Initial conditions (initial displacement and initial velocity):

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x), \qquad x \in \mathbb{R}. \tag{2}$$

- $u(x,t)$ is the displacement of the point x at time t; $k > 0$ is the wave speed.
- We assume $f \in C^2(\mathbb{R})$ and $g \in C^1(\mathbb{R})$.
- (1)–(2) is a Cauchy problem (initial value problem) for the wave equation.
- No boundary conditions are needed, since the string has no ends.



Recall: Canonical Form and General Solution

- The wave equation is hyperbolic with characteristics $x \pm kt =$ const.
- With $\xi = x + kt$, $\eta = x - kt$ it reduces to the canonical form

$$u_{\xi\eta} = 0. \tag{3}$$

- Integrating, the general solution of (3) is $u = \phi(\xi) + \psi(\eta)$, i.e.

$$u(x,t) = \phi(x + kt) + \psi(x - kt), \tag{4}$$

  where ϕ and ψ are arbitrary twice differentiable functions.
- We now choose ϕ and ψ to satisfy the initial conditions (2).



Imposing the Initial Conditions

- From (4), $u_t(x,t) = k\,\phi'(x+kt) - k\,\psi'(x-kt)$.
- Putting $t = 0$ in u and u_t and using (2):

$$u(x,0) = \phi(x) + \psi(x) = f(x), \tag{5}$$

$$u_t(x,0) = k\big[\phi'(x) - \psi'(x)\big] = g(x). \tag{6}$$

- Integrate (6) from 0 to x:

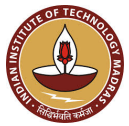
$$k\int_0^x \big[\phi'(s) - \psi'(s)\big]\,ds = \int_0^x g(s)\,ds.$$

- Hence $k\big[\phi(x) - \psi(x)\big] - k\big[\phi(0) - \psi(0)\big] = \displaystyle\int_0^x g(s)\,ds.$



Solving for ϕ and ψ

- Thus
$$\phi(x) - \psi(x) = \frac{1}{k}\int_0^x g(s)\,ds + \phi(0) - \psi(0). \tag{7}$$
- Adding (5) and (7):
$$\phi(x) = \frac{1}{2}f(x) + \frac{1}{2k}\int_0^x g(s)\,ds + \frac{1}{2}\big[\phi(0) - \psi(0)\big]. \tag{8}$$
- Subtracting (7) from (5):
$$\psi(x) = \frac{1}{2}f(x) - \frac{1}{2k}\int_0^x g(s)\,ds - \frac{1}{2}\big[\phi(0) - \psi(0)\big]. \tag{9}$$
- These hold for every real argument, so we may replace x by $x + kt$ and $x - kt$.



D'Alembert's Formula

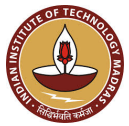
- From (4), (8) and (9):

$$u(x,t) = \tfrac{1}{2}f(x+kt)+\tfrac{1}{2k}\int_0^{x+kt} g(s)\,ds+\tfrac{1}{2}f(x-kt)-\tfrac{1}{2k}\int_0^{x-kt} g(s)\,ds$$

- The constants $\pm\frac{1}{2}[\phi(0)-\psi(0)]$ cancel.
- Combining the two integrals:

$$\boxed{u(x,t) = \frac{1}{2}\big[f(x+kt)+f(x-kt)\big] + \frac{1}{2k}\int_{x-kt}^{x+kt} g(s)\,ds}. \tag{10}$$

- This is D'Alembert's solution of the Cauchy problem (1)–(2).
- Since f, g are defined on all of $\mathbb{R}$, $u(x,t)$ is well defined for every x and t.



Verification of D'Alembert's Formula

- Let u be given by (10); by the Leibniz rule,

$$u_t = \frac{k}{2}\big[f'(x+kt) - f'(x-kt)\big] + \frac{1}{2}\big[g(x+kt) + g(x-kt)\big].$$

- At $t = 0$: $u(x,0) = f(x)$ and $u_t(x,0) = 0 + g(x) = g(x)$.
- Differentiating again:

$$u_{tt} = \frac{k^2}{2}\big[f''(x+kt) + f''(x-kt)\big] + \frac{k}{2}\big[g'(x+kt) - g'(x-kt)\big].$$

- Similarly

$$u_{xx} = \frac{1}{2}\big[f''(x+kt) + f''(x-kt)\big] + \frac{1}{2k}\big[g'(x+kt) - g'(x-kt)\big].$$

- Hence $u_{tt} = k^2 u_{xx}$, and (10) solves the Cauchy problem.



Well-posedness

- A problem is well posed if a solution exists, is unique, and depends continuously on the data.
- Existence: D'Alembert's formula (10) gives a solution when $f \in C^2$, $g \in C^1$.
- Uniqueness: every solution has the form (4), and (5)–(9) force formula (10).
- Stability: let u_1, u_2 correspond to data (f_1, g_1), (f_2, g_2) with

$$|f_1 - f_2| < \delta, \qquad |g_1 - g_2| < \delta \quad \text{on } \mathbb{R}.$$

- Then from (10), for $0 \le t \le T$:

$$|u_1 - u_2| \le \tfrac{1}{2}(\delta + \delta) + \tfrac{1}{2k}\,(2kt)\,\delta = \delta\,(1 + t) \le \delta\,(1 + T).$$

- Small changes in the data produce small changes in the solution on finite time intervals.



Forward and Backward Waves

- The general solution is $u(x,t) = \phi(x+kt) + \psi(x-kt)$.
- Replace x by $x+k$ and t by $t+1$: $\psi(x-kt)$ is unchanged.
- So $\psi(x-kt)$ is a profile moving to the right with speed k (forward wave).
- Replace x by $x-k$ and t by $t+1$: $\phi(x+kt)$ is unchanged.
- So $\phi(x+kt)$ is a profile moving to the left with speed k (backward wave).
- The solution is a superposition of two waves travelling in opposite directions with the same speed k.
- The shapes do not change while travelling: there is no dispersion.



Special Case: Zero Initial Velocity ($g = 0$)

- If the string is released from rest, $g \equiv 0$ and (10) becomes

$$u(x,t) = \frac{1}{2}\big[f(x+kt) + f(x-kt)\big].$$

- The initial profile f splits into two halves of equal shape.
- One half, $\frac{1}{2}f(x-kt)$, moves to the right with speed k.
- The other half, $\frac{1}{2}f(x+kt)$, moves to the left with speed k.
- Each half has amplitude equal to half the initial amplitude.
- Example: a string plucked at the middle of a short segment and released.
- The value $u(x,t)$ is the average of the initial displacements at $x-kt$ and $x+kt$.



Special Case: Zero Initial Displacement ($f = 0$)

- If $f \equiv 0$ (the string is struck, not displaced), then

$$u(x,t) = \frac{1}{2k} \int_{x-kt}^{x+kt} g(s)\, ds.$$

- Write this as

$$u(x,t) = t \cdot \left[\frac{1}{2kt} \int_{x-kt}^{x+kt} g(s)\, ds \right].$$

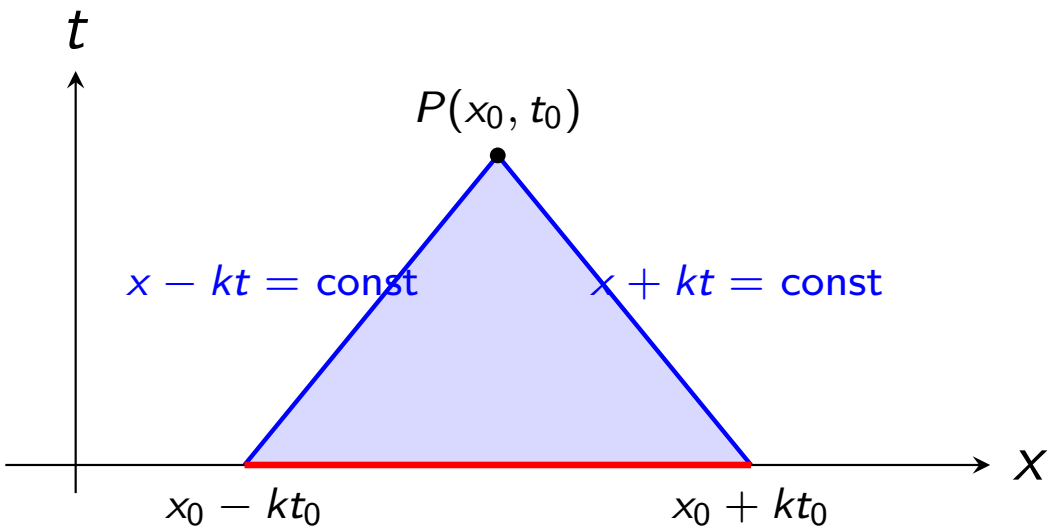
- The bracket is the average initial velocity over $[x - kt,\ x + kt]$.
- So in general, D'Alembert's solution reads.
- $u(x,t) =$ (average of the initial deflection at the end points of $[x - kt, x + kt]$).
- $+\ t\times$ (average initial velocity over $[x - kt, x + kt]$).



Domain of Dependence

- By (10), $u(x_0, t_0)$ uses f only at $x_0 \pm kt_0$ and g only on $[x_0 - kt_0,\ x_0 + kt_0]$.
- This interval is the domain of dependence of the point (x_0, t_0):

$$\{\, x : |x - x_0| \leq kt_0 \,\}.$$



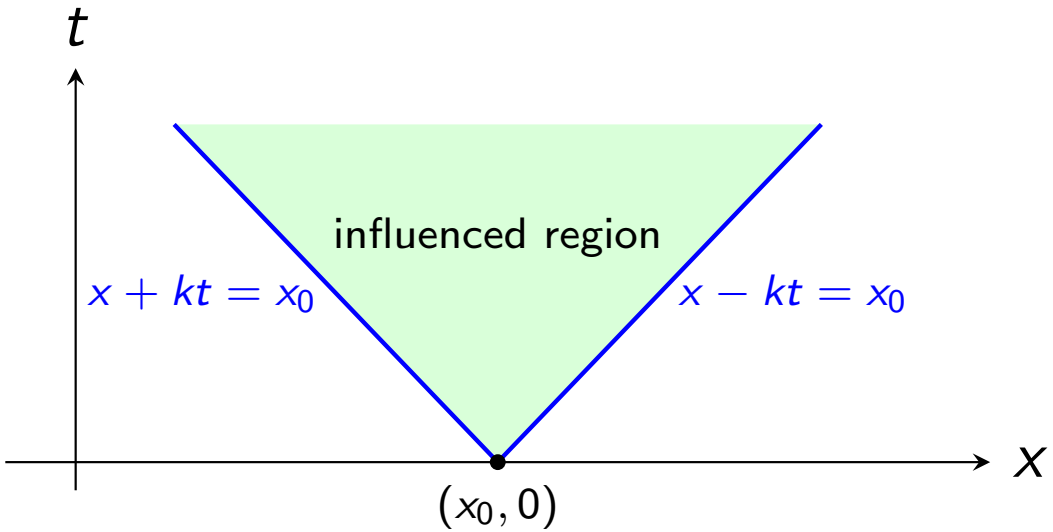
- It is cut out by the two characteristics through P (the characteristic triangle).

Domain of Influence

- Conversely, the initial values at a point $(x_0, 0)$ affect u only in the set

$$\{\,(x,t) : |x - x_0| \le kt,\ \ t \ge 0\,\}.$$

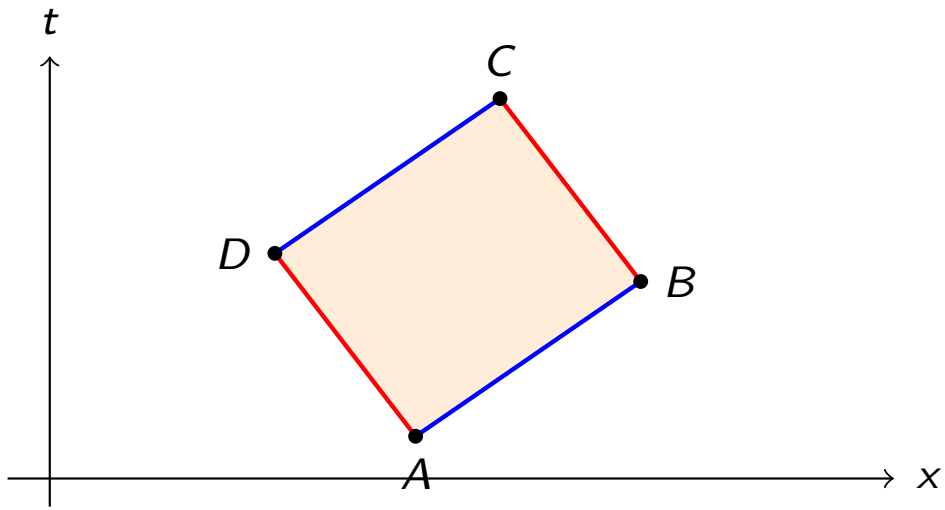
- This is the domain of influence of the point $(x_0, 0)$.



- A disturbance at x_0 reaches the point x only after time $|x - x_0|/k$: finite speed of propagation.

Characteristic Parallelogram Property

- Let $ABCD$ be a parallelogram bounded by characteristics, with A, C opposite vertices.



- Every solution $u = \phi(x + kt) + \psi(x - kt)$ satisfies

$$u(A) + u(C) = u(B) + u(D).$$

- Reason: $\xi = x + kt$ and $\eta = x - kt$ take only two values each on the vertices.
- Useful for computing u near boundaries from known values.

Example 1: $f(x) = e^{-x^2}$, $g(x) = \cos x$

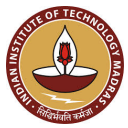
- Solve $u_{tt} = k^2 u_{xx}$, $x \in \mathbb{R}$, $t > 0$, with $u(x,0) = e^{-x^2}$, $u_t(x,0) = \cos x$.
- By D'Alembert's formula,

$$u(x,t) = \frac{1}{2}\left[e^{-(x+kt)^2} + e^{-(x-kt)^2}\right] + \frac{1}{2k}\int_{x-kt}^{x+kt} \cos s \, ds.$$

- The integral equals $\sin(x+kt) - \sin(x-kt) = 2\cos x \, \sin kt$.
- Hence

$$u(x,t) = \frac{1}{2}\left[e^{-(x+kt)^2} + e^{-(x-kt)^2}\right] + \frac{1}{k}\cos x \, \sin kt.$$

- Check: $u(x,0) = e^{-x^2}$ and $u_t(x,0) = \cos x$.



Example 2: $f = 0, \quad g(x) = \dfrac{1}{1+x^2}$

- Here only the initial velocity is non-zero.
- By D'Alembert's formula,

$$u(x,t) = \frac{1}{2k}\int_{x-kt}^{x+kt} \frac{ds}{1+s^2} = \frac{1}{2k}\Big[\tan^{-1}(x+kt) - \tan^{-1}(x-kt)\Big].$$

- At a fixed x, as $t \to \infty$: $\quad u(x,t) \to \dfrac{1}{2k}\left[\dfrac{\pi}{2} + \dfrac{\pi}{2}\right] = \dfrac{\pi}{2k}.$
- The string is permanently displaced: $\frac{\pi}{2k} = \frac{1}{2k}\int_{-\infty}^{\infty} g(s)\,ds.$
- Contrast with $g = 0$, where a localized pulse leaves every fixed point eventually at rest.



Example 3: Piecewise Initial Velocity

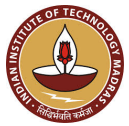
- Solve $u_{tt} = 4u_{xx}$, $x \in \mathbb{R}$, $t > 0$, with

$$u(x,0) = 0, \qquad u_t(x,0) = g(x) = \begin{cases} 1, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}.$$

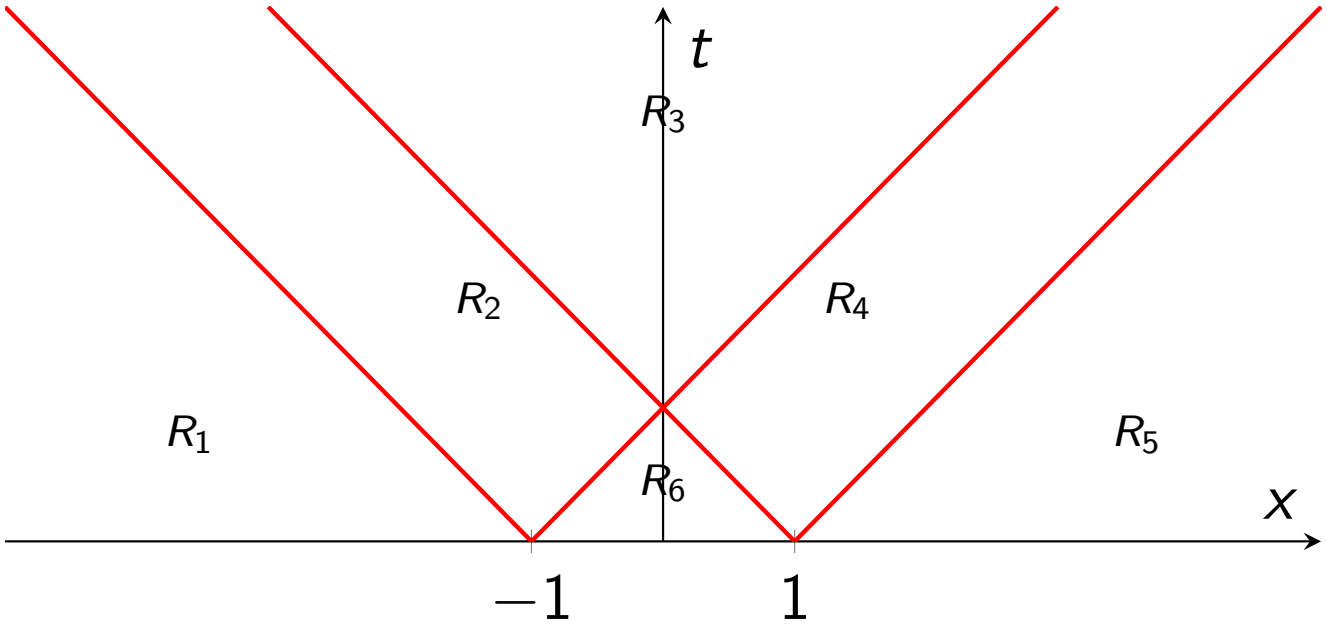
- Here $k = 2$ and $f = 0$, so

$$u(x,t) = \frac{1}{4}\int_{x-2t}^{x+2t} g(s)\,ds = \frac{1}{4}\Big|\,[x-2t,\ x+2t] \cap (-1,1)\,\Big|.$$

- ($|\cdot|$ denotes the length of the interval.).
- The answer depends on where $x \pm 2t$ lie relative to ± 1.
- The characteristics $x \pm 2t = \pm 1$ through $(\pm 1, 0)$ divide the upper half plane into six regions.



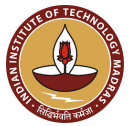
Example 3: The Six Regions



- The lines are $x+2t=-1$, $x-2t=-1$, $x+2t=1$, $x-2t=1$.
- R_6 is the characteristic triangle over the interval $(-1,1)$.

Example 3: The Solution

- R_1: $x + 2t \leq -1$ (interval left of $(-1, 1)$): $u = 0$.
- R_5: $x - 2t \geq 1$ (interval right of $(-1, 1)$): $u = 0$.
- R_6: $-1 \leq x - 2t,\ x + 2t \leq 1$ (inside): $u = \frac{1}{4}(4t) = t$.
- R_2: $x - 2t \leq -1 \leq x + 2t \leq 1$: $u = \frac{1}{4}(x + 2t + 1)$.
- R_4: $-1 \leq x - 2t \leq 1 \leq x + 2t$: $u = \frac{1}{4}(1 - x + 2t)$.
- R_3: $x - 2t \leq -1,\ x + 2t \geq 1$ (interval covers $(-1, 1)$): $u = \frac{1}{4} \cdot 2 = \frac{1}{2}$.
- u is continuous across the lines but its derivatives jump there.
- Discontinuities in g travel along the characteristics through $x = \pm 1$.



Example 4: Polynomial Data

- Solve $u_{tt} = k^2 u_{xx}$ with $u(x,0) = x^2$, $u_t(x,0) = x$.
- $\frac{1}{2}\left[(x+kt)^2 + (x-kt)^2\right] = x^2 + k^2t^2$.
- $\dfrac{1}{2k}\displaystyle\int_{x-kt}^{x+kt} s\,ds = \dfrac{1}{4k}\left[(x+kt)^2 - (x-kt)^2\right] = xt.$
- Hence
$$u(x,t) = x^2 + k^2t^2 + xt.$$
- Check: $u_{tt} = 2k^2$, $u_{xx} = 2$; $u(x,0) = x^2$, $u_t(x,0) = x$.
- For polynomial data, D'Alembert's formula gives polynomial solutions.

